

1. Prerequisite Study Guide (Kya Padhna Hai?)

Is assignment ko attempt karne se pehle sabhi team members ko niche diye gaye concepts ko ache se revise kar lena hai:

- **Exception Handling Mechanics:** try, except, else, aur finally blocks ka execution flow.
- **Error Types & Identification:** Common exceptions jaise ValueError, ZeroDivisionError, TypeError, aur khas karke FileNotFoundError (vs FileExistsError).
- **File Handling (File I/O):** Safe file opening using with open(), file modes (Read "r", Write "w", Append "a"), aur file reading/writing methods.

2. Industry Relevance (Kyon aur Kahan Kaam Aayega?)

- **Crash-Proof Production Systems:** Real-world applications aur backend servers mein unexpected errors ke aane par application ko crash hone se bachane ke liye Exception Handling zaroori hai.
- **AI/ML Data Pipelines & Checkpoints:** Machine learning models (jaise PyTorch/TensorFlow) ki training ke dauran hazaron data files process hoti hain. Agar 1-2 files corrupt ya missing hon, toh try-except 10 ghante ki training ko bich mein rukne se bachata hai. Iske alawa, with open() model weights (checkpoints) save karne aur server logs maintain karne mein har jagah use hota hai.

3. Short & Focused Assignment Tasks

Task 1: Safe Division Utility (Exception Handling)

Requirement: Ek function safe_calculator(num1, num2) banayein jo do numbers ko divide kare aur errors ko gracefully handle kare.

- try block ke andar division num1 / num2 perform karein.
- except ZeroDivisionError catch karke return karein: "Error: Zero se divide nahi kar sakte!"
- except TypeError catch karke return karein: "Error: Dono inputs numbers (int/float) hone chahiye!"

Task 2: Devdoots Activity Logger (File I/O)

Requirement: Ek script likhein jo team activity ko local file mein log (append) aur read kare.

- Append mode ("a") ka use karke devdoots_activity.txt file mein apni activity line write karein: "[Member Name]: Practiced Exception Handling and File I/O\n".
- Read mode ("r") mein us file ko safely open karke read karein. Read operation ko try-except (FileNotFoundError) block mein rakhein aur content print karein.

4. GitHub Submission Rules & Workflow (Single File Push)

Main branch par direct commit karna prohibited hai. Single file push karne ke liye niche diya gaya standard TPM workflow follow karein:

Step	Git Command	Description
1. Sync Main Branch	git checkout main git pull origin main	Remote main branch se local repository ko sync karein.
2. Create Feature Branch	git checkout -b user/yourname-assign4	Apni alag branch banayein (e.g., user/divyansh-assign4).
3. Create Assignment File	File Name: divyansh_assign4.py	Apne folder mein sirf apni single python file banayein.

Step	Git Command	Description
4. Stage & Push Single File	<pre>git add divyansh_assign4.py git commit -m "[HW-4] Added Exception Handling and File IO" git push origin user/yourname-assign4</pre>	Important: Direct git add . mat karna. Sirf apni single file stage karke push karein.
5. Raise Pull Request	GitHub Portal → Compare & pull request	GitHub par jaakar Pull Request (PR) raise karein.